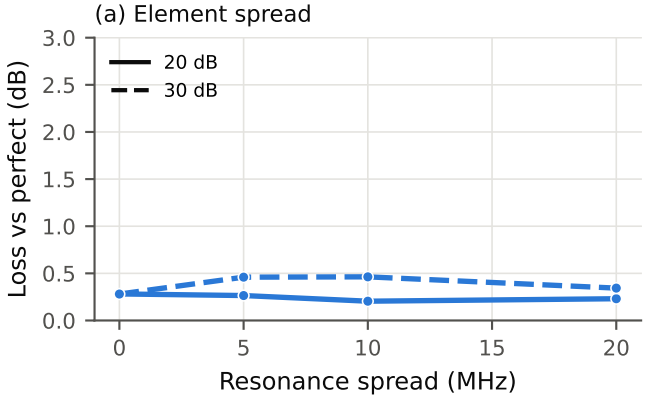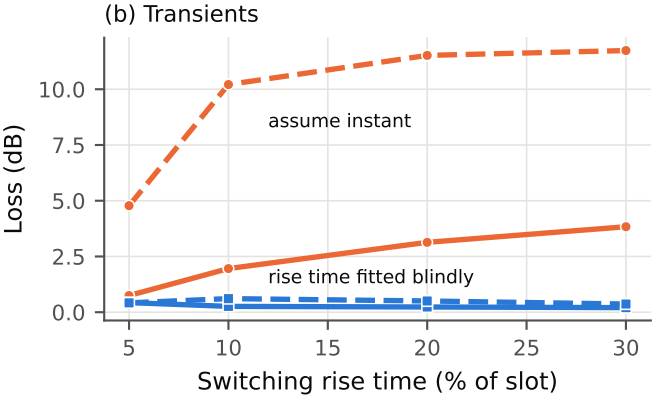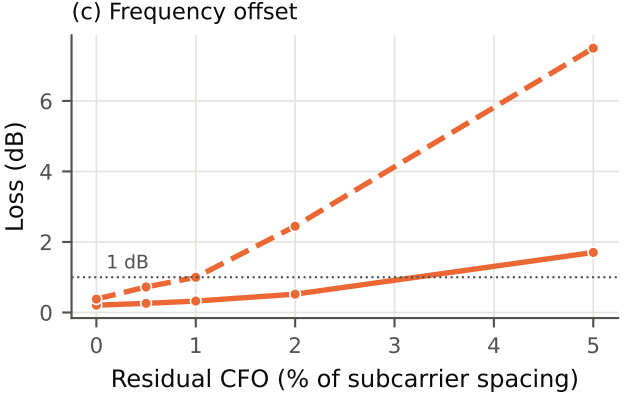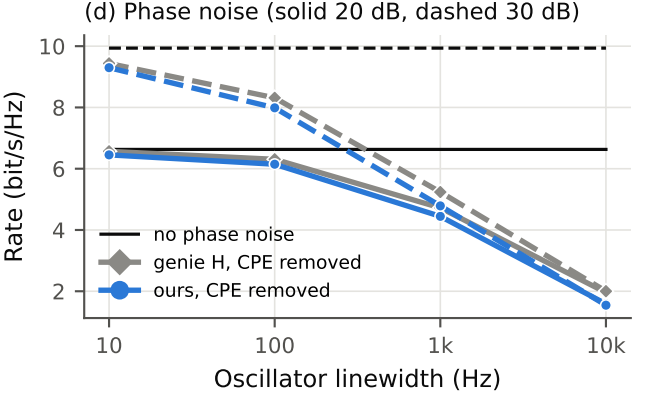Spread is harmless; transients are fixed by fitting the rise time;
CFO must stay below ~1% and oscillator linewidth near 100 Hz